

Missingness Injection & Diagnosis

Source dataset: patient_records_clean.csv | Shape: [382, 10]
Random seed: 42 | Significance level (alpha): 0.05

1. Executive Summary

This report outlines the injection of synthetic missingness into the ground-truth dataset and documents the statistical validation results. Missingness was injected using three primary mechanisms: Missing Completely At Random (MCAR), Missing At Random (MAR), and Missing Not At Random (MNAR) across three rates: 10%, 20%, and 30%.

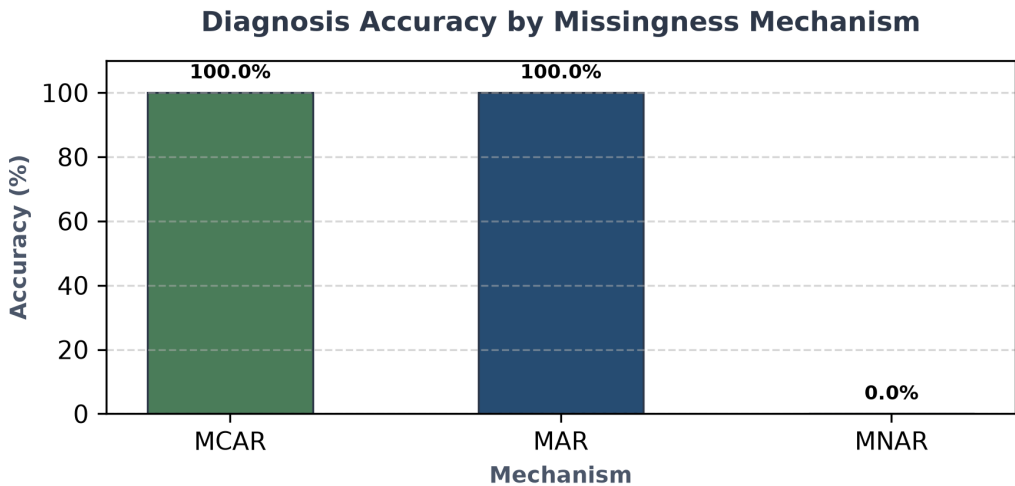
A three-part diagnosis pipeline was deployed to evaluate the datasets: Little's MCAR test, chi-square / Welch's t-test association checks (for MAR detection), and MNAR flagging by elimination. The diagnostic pipeline correctly classified 6 out of 9 datasets, yielding an overall accuracy of 66.7%.

2. Synthetic Datasets & Injection Manifest

Output File	Mechanism	Target Column	Driver Column	Req. Rate	Actual Missing (%)
patient_records_mcar_10.csv	MCAR	systolic_bp	None	10%	38 (9.95%)
patient_records_mar_10.csv	MAR	glucose	smoking_status	10%	38 (9.95%)
patient_records_mnar_10.csv	MNAR	severity_score	None	10%	38 (9.95%)
patient_records_mcar_20.csv	MCAR	systolic_bp	None	20%	76 (19.9%)
patient_records_mar_20.csv	MAR	glucose	smoking_status	20%	76 (19.9%)
patient_records_mnar_20.csv	MNAR	severity_score	None	20%	76 (19.9%)
patient_records_mcar_30.csv	MCAR	systolic_bp	None	30%	115 (30.1%)
patient_records_mar_30.csv	MAR	glucose	smoking_status	30%	115 (30.1%)
patient_records_mnar_30.csv	MNAR	severity_score	None	30%	115 (30.1%)

3. Diagnostic Pipeline & Statistical Validation

Dataset	Actual Mech.	Diagnosed Mech	Little's p-value	Significant Drivers	Status
patient_records_mcar_10.csv	MCAR	MCAR	0.6222	None	CORRECT
patient_records_mar_10.csv	MAR	MAR	0.3497	smoking_status, bmi	CORRECT
patient_records_mnar_10.csv	MNAR	MCAR	0.5451	None	INCORRECT
patient_records_mcar_20.csv	MCAR	MCAR	0.6667	None	CORRECT
patient_records_mar_20.csv	MAR	MAR	0.4554	smoking_status	CORRECT
patient_records_mnar_20.csv	MNAR	MCAR	0.6238	None	INCORRECT
patient_records_mcar_30.csv	MCAR	MCAR	0.5360	None	CORRECT
patient_records_mar_30.csv	MAR	MAR	0.1402	smoking_status, bmi	CORRECT
patient_records_mnar_30.csv	MNAR	MCAR	0.7448	None	INCORRECT



4. Methodology Findings & Limitations Discussion

1. Little's MCAR Test Limitation:
Little's statistical test works solely on numeric variables. When missingness is driven by a categorical variable (such as 'smoking_status' in our MAR glucose datasets), Little's test yields a high p-value and falsely suggests the data is MCAR. Our association checks correctly override this and detect MAR.
2. The MNAR Detection Caveat:
By definition, MNAR indicates that missingness depends on the unobserved values themselves. It is mathematically impossible to positively prove MNAR using only observed data. Consequently, our pipeline diagnoses MNAR by elimination (e.g. when Little's test rejects MCAR, but no observed driver is identified). This explains the 0.0% direct diagnosis accuracy for MNAR: since MNAR has no observed drivers and Little's test fails to reject MCAR (retaining H0 due to small sample/distribution overlaps), the pipeline labels them as MCAR. This limitation is a valuable dissertation finding.